



DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

**An Explainable AI Framework for
Ear Disease Classification Using Otoscopic Images.**

submitted to
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Submitted by

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Bachelor of Engineering in Information Science & Engineering

Under the guidance of

Dr D. Evangeline
Assistant Professor

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

RAMAIAH INSTITUTE OF TECHNOLOGY
(Autonomous Institute, Affiliated to VTU)



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CERTIFICATE

This is to certify that the project work titled **An Explainable AI Framework for Ear Disease Classification Using Otoscopic Images** is a bonafide work carried out by Chinmayi Suresh Shetty (1MS23IS034) , Hrishika R (1MS23IS044) , S Yashaswini (1MS23IS107) and Bhoomikumari A (1MS24IS404) in partial fulfillment for the course of Bachelor of Engineering in Information Science and Engineering during the year 2026. The Project report has been approved as it satisfies the academic requirements with respect to the project work. To the best of our understanding, the work submitted in this report has not been submitted anywhere.

Signature of the Guide

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- 2.

DECLARATION

We students of sixth semester BE, Dept. of Information Science and Engineering, Ramaiah Institute of Technology, Bangalore, hereby declare that the project entitled “**An Explainable AI Framework for Classification of Ear Diseases Using Otoscopic Images**”, thesis completed and written by us under the guidance of Dr. D. Evangeline, Dept. of ISE, Bangalore.

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ABSTRACT

Ear disease is a common health issue that affects individuals of any age group and can cause hearing-related complications if left undiagnosed. In hospitals, diagnosis is commonly done through symptom analysis and ear examination with an instrument such as an otoscope. This process can be slow and heavily reliant on clinical experience. With the increasing use of Artificial Intelligence and Machine Learning, automated systems are being utilized to aid in medical diagnosis and decision-making. The primary objective of this project is to develop an AI system that can classify ear diseases by considering patient symptoms and otoscopic ear scans. Rather than using only one input, the system incorporates structured symptom information and medical image analysis to enhance accuracy in making predictions. The analysis of symptom data involves the use of Random Forest model and image classification using Vision Transformer (ViT) model. Using the late fusion technique, both models' outputs are combined to create their final predictions. The system's clarity and dependability are further enhanced by the inclusion of Explainable AI techniques. SHAP is utilized to identify the symptoms that are most relevant to the prediction, while Grad-CAM highlights important areas in ear imaging used for classification. Additionally, an LLM is present to generate uncomplicated clinical explanations based on the prediction outcomes and explainability outputs. Upon UCI Otoscopic Image dataset and synthetic symptom information, the proposed methodology gives 98% accuracy, 98% precision, 98% recall and 98% F1-score.

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1.INTRODUCTION

1.1 General Introduction

Ear diseases are common medical conditions that affect individuals across all age groups and can lead to hearing impairment, discomfort, and long-term complications if not diagnosed at an early stage [1]. Traditional diagnosis methods involve clinical examination using an otoscope along with patient-reported symptoms [2]. However, these methods are often time-consuming and require expert interpretation.

A significant aspect of medical diagnosis systems has led to rapid progress made by Artificial Intelligence (AI) and Machine Learning (ML). In the healthcare industry, deep learning techniques have been successful in resolving image classification problems [4][5]. Using attention mechanisms, Vision Transformer (ViT) models can capture global relationships in images, which has rendered traditional Convolutional Neural Networks (CNN-based) obsolete. The use of multimodal learning, which involves merging medical images with patient symptoms to improve diagnostic accuracy, has become a significant advancement in addition to image analysis. [2][6][12]. Furthermore, Explainable AI (XAI) techniques like SHAP and Grad-CAM aid in comprehending model decisions, resulting in increased trust and transparency [10]. Using a Vision Transformer (ViT) [3][11], this project seeks to develop an AI system that can integrate symptom data and images, with the aim of producing clear output that conveys multiple modalities. Furthermore, a Large Language Model (LLM) is utilized by the system to produce clinical explanations that are easily understood by humans, improving the interpretability of these predictions and making them valuable in real-life healthcare settings.

1.2 Problem Statement

Currently, existing AI-based systems for ear disease detection are mostly based on image classification and do not consider patient complaints[4][5]. As a result, many existing systems cannot provide full knowledge about the patient's condition. Furthermore, numerous deep learning models utilized for medical image analysis are challenging to interpret, rendering their predictions unclear in the context of clinical applications[10].

Traditional Convolutional Neural Network (CNN) models prioritize local feature extraction over capturing broader relationships in medical images[3][11]. A significant flaw in most current systems is that they lack clinically comprehensible explanations, which makes them less practical for use in healthcare settings.

1.3 Objectives of the Project

The main objectives of this project are:

- 1) To develop a multimodal AI system integrating patient symptoms and otoscopic images.
- 2) To implement Vision Transformer (ViT) for improved image classification and apply Random Forest for symptom-based disease prediction.
- 3) To perform multimodal late fusion to enhance diagnostic accuracy and incorporate Explainable AI techniques such as SHAP and Grad-CAM.
- 4) To generate human-readable clinical explanations using Large Language Models (LLMs) and evaluate the developed system using relevant performance measures.

1.4 Project Deliverables

The project provides a full-scale multimodal ear disease classification system that incorporates both symptoms and images of the otoscope (ear) to aid in diagnosis. Among the features of the system are its model, which uses Vision Transformers, and its module, called Random Forest, for identifying symptoms. Additionally, there is another module available to aid in this effort. The final prediction is based on the outputs of both models being combined by a multimodal fusion mechanism. SHAP and Grad-CAM techniques are utilized by the framework to provide outputs that enhance transparency and interpretability. Furthermore, there is an LLM module that aims to produce clinical explanations in human-readable language using the prediction results and explainability outputs. The project's user interface facilitates interaction with the system and simplifies the visualization of predictions and explanations.

1.4.1 Deliverable Development and Review Process

The process of developing the project is structured around requirement analysis and literature survey. It is then developed and executed through a series of stages i.e., symptom-based prediction, image classification, multimodal fusion with explanation integration, and testing. The complete system is formed by assembling individual modules. Accuracy, efficiency and reliability are ensured through continuous evaluation and testing. Each stage contains feedback to improve the system's performance, interpretability and usable capabilities.

1.5 Current Scope

A multimodal AI system is being developed by the project, which can analyze patient symptoms and classify ear diseases using otoscopic images. This is supported by the system which can be used to classify images using Vision Transformers, use machine learning for symptom prediction purposes, perform multimodal fusion tasks such as SHAP and Grad-CAM and generate clinical explanations using LLM methods. The primary purpose of the implementation is to serve as a demonstration and academic tool for publicly accessible datasets.

1.6 Future Scope

There exist multiple possibilities for extending the project to enhance its practicality and effectiveness. Future developments may involve integrating with real-time clinical and hospital systems to provide hands-on healthcare support, or potentially deploying on cloud platforms to increase scale and accessibility. Multimodal analysis can be improved by incorporating additional medical information, such as audiometry reports, tympanometry data and other diagnostic data. Using larger and more diverse datasets can improve the generalization of models by providing greater accuracy. This module is based on the LLM explanation, which can be further developed using domain-specific medical knowledge to produce more clinically relevant interpretations. Additionally, the framework can be adapted to form a mobile diagnostic application for faster diagnosis and remote healthcare support.

2. PROJECT ORGANIZATION

2.1 Software Process Model

It is developed using an Agile software process model with emphasis on iterative development, continuous testing and regular feedback. Every new idea is iterated and added by us to improve the overall idea of our project. Each phase of the project concentrates on particular components, such as symptom analysis, image classification through Vision Transformer (ViT), multimodal fusion, explainability, and clinical explanation generation using a Large Language Model (LLM). Adaptive adaptation allows for flexibility in the implementation of changes and improvements. Each stage involves planning, design and implementation, testing and early detection of errors to ensure corrective action.

It helped us to compare different possibilities of our project and to improve our work.

2.2 Roles and Responsibilities

Table 2.2 Roles and Responsibilities

Sl. No	Name	USN	Role	Responsibilities
1.	CHINMAYI SHETTY	1MS23IS034	Frontend & UI Development.	Designed an interface to handle user input for symptoms and images, displayed predictions and necessary explanations.
2.	HRISHIKA R	1MS23IS044	System Integration.	Designed the overall architecture, integrated ViT, Random Forest, fusion and LLM modules.
3.	S YASHASWINI	1MS23IS107	Model & Explainability Lead	Implemented Vision Transformer, SHAP, Grad-CAM and analyzed outputs
4.	BHOOMI KUMARI A	1MS24IS404	Backend & Data Processing Lead	Handled preprocessing of images, symptom data, dataset preparation, symptom modeling and fusion logic.

2.3 Finance / Budget Estimation

It is a low-cost project that relies heavily on open-source tools and frameworks.

- Vision Transformer (ViT): Open-source platforms (HuggingFace / PyTorch) .
- Random Forest Model: Implemented using Scikit-learn (free).
- SHAP & Grad-CAM: Open-source libraries.
- LLM APIs (optional): Cost depends on usage (if using external APIs) .
- Development Tools: Python, Gradio(free) .
- Hardware: Standard system with moderate RAM and GPU (optional) .

The project is feasible with a small budget and can be carried out academically.

2.4 Sustainable Development Goals (SDGs)

This project aligns with the following SDGs:

- **SDG 3: Good Health and Well-being**
It improves early detection of ear diseases using AI frameworks.
- **SDG 4: Quality Education**
It helps in understanding AI-based medical systems and learning healthcare applications.
- **SDG 9: Industry, Innovation, and Infrastructure**
It serves to promote innovation through multimodal AI and explainable systems.
- **SDG 17: Partnerships for the Goals**
It uses open-source tools and collaborative research contributions.

3. LITERATURE SURVEY

3.1 Introduction

The literature review highlights the progress achieved in ear disease detection using artificial intelligence and multimodal learning methods [1][2]. The use of deep learning has expanded to include Convolutional Neural Networks (CNNs), Vision Transformers (ViT) and hybrid architectures for medical image classification tasks [3][4][11]. The significance of combining various data modalities, such as otoscopic images, symptoms, and audiometry data in improving diagnostic accuracy and clinical reliability [2][6][12], has been highlighted by recent studies. Furthermore, medical systems are increasingly incorporating Explainable AI techniques to enhance the transparency and comprehensibility of medical decisions. For the development of the proposed multimodal explainable AI framework, this survey examines the different methodologies, datasets, observations, and limitations associated with existing systems.

In the work [1], the proposed approach employs an ensemble model based on CNN with Bayesian optimization for better classification of ear diseases. A public otoscopic image dataset of 880 images was utilized in the study and GradCAM++ was employed for explanation. Performance was enhanced through hyperparameter tuning, as demonstrated. Although it incorporated image-based classification, the system did not incorporate multimodal learning or patient symptoms.

The work [2] combined otoscopic image data with tympanometry data. A more accurate diagnosis was achieved with multimodal systems than with unimodal ones. The classification performance was improved, but the system lacked the ability to explain it or to broaden the use of multimodal methods. In [3] extraction of features from otoscopic images using Vision Transformer (ViT) was reported. The use of kNN, SVM, and Random Forest was employed to implement machine learning classifiers for final classification. Using ViT with the neural network kNN, this study was almost completely accurate and showed that attention is learned. Nevertheless, the work was limited to data in terms of image quality and did not incorporate any features for explaining them. The work [4] proposed an algorithm for otoscopic images that uses deep learning and CNN. Using publicly available ear image datasets, the model was constructed with a high level of classification precision. The framework's performance was satisfactory, but it failed to include multimodal learning and provide explanations for AI integration.

In paper [5] researchers used Deep Convolutional Neural Networks (DCNNs) to automatically detect middle ear abnormalities for disease detection using deep learning techniques. [5]. Approximately 96-98% of the model's accuracy in detecting abnormalities in the middle ear was achieved. This framework lacked the necessary image-based diagnosis, nor could it support analysis or interpretation of symptoms.

In [6] they proposed a multimodal fusion model that combined EfficientNetB7 image elements with hearing test data. Diagnostic performance was found to improve by integrating audiometry information with image data, according to the study. However, the system faced challenges in situations where hearing loss was not purely auditory. The work [7] utilized transfer learning

models like ResNet50 and EfficientNet. Accuracy was significantly higher when transferred learning than when training from scratch. The framework did not provide sufficient support for multimodal symptom integration and explainability. In paper [8], an edge computing device was used to implement an EfficientNetB5-based CNN model for real-time classification. This setup maintained consistent performance on hardware with low resources. In spite of this, it failed to consider patient symptoms and underwent a slight performance decline due to limited resources.

The work [9] utilized image embeddings extracted through Inception V3, as well as SVM, kNN, and neural networks for classification. This model was effective in handling large amounts of CT scan images. The work only covered CT images and lacked multimodality and the ability to describe AI. The work [10] combined CNN models with Grad-CAM visualization techniques to provide a clear explanation. By highlighting image areas that are responsible for making predictions, the framework increased transparency. However, the system was only able to diagnose using images and not multimodal learning.

The research in [11] paper proposed a hybrid architecture that incorporates both. The model's ability to extract features and classify them with precision outperformed those of conventional methods. Nevertheless, the framework mandated extensive datasets and inadequate clinical validation. The work [12] employed CNN-based image classification and Random Forest symptom analysis. Diagnostic accuracy was enhanced by utilizing the multimodal system, which combined image data with symptoms. The new system. But the dataset was too small, and there was not enough implementation of explainability techniques.

Table 3.2 Literature Survey Summary

Work	Methodology	Datasets	Observation	Limitations
[1]	Convolutional Neural Network (CNN) integrated with Bayesian Optimization for hyperparameter tuning. GradCAM++ was used for explainability and visualization of predictions.	Public otoscopic ear image dataset containing 880 images.	Improved classification accuracy by 5% through parameter optimization and explainability integration.	Does not integrate patient symptoms and lacks hybrid learning design.
[2]	Combined otoscopic image data and tympanometry using Support Vector Machine (SVM) models for multimodal disease diagnosis.	Clinical dataset containing matched video otoscopy and tympanometry data.	Multimodal approach improved diagnostic accuracy providing 98% accuracy level.	Limited interpretability and lacks advanced explainability mechanisms.

Work & Citation	Methodology	Datasets	Observation	Limitations
[3]	Vision Transformer (ViT) used as feature extractor with machine learning classifiers such as kNN, SVM, and Random Forest for classification.	Otoscopic dataset containing 880 eardrum images categorized into CSOM, earwax, myringosclerosis, and normal conditions.	ViT combined with kNN achieved nearly 99% classification accuracy demonstrating strong performance.	Focused only on image-based diagnosis and lacked explainability integration.
[4]	CNN-based deep learning architecture for multiclass otoscopic image classification.	Public multiclass otoscopic ear image dataset.	Achieved high classification accuracy using deep learning approaches.	Limited interpretability and unimodal image-based diagnosis.
[5]	Deep Convolutional Neural Networks (DCNNs) for automated middle ear abnormality detection.	Labeled otoscopic image dataset containing normal and pathological cases.	Achieved high detection accuracy (96–98%) and effectively extracted disease-specific visual features.	No patient symptom integration and no Explainable AI framework for interpretation.
[6]	EfficientNet-B7 image model combined with MLP hearing test analysis using multimodal fusion techniques.	1,480 endoscopic images paired with Pure-Tone Audiometry (PTA) data.	Fusion of hearing tests and image data improved identification of ear diseases.	Difficult to diagnose effectively when only image datasets are available.
[7]	Transfer learning using ResNet50 and EfficientNet for multiclass ear disease classification.	Public Ear Imagery Database containing 880 images.	Transfer learning significantly improved classification accuracy compared to training from scratch.	Lacks symptom integration and Explainable AI support.

[8]	EfficientNetB5 CNN deployed on Rockchip RK3566 edge device with data augmentation and Group Normalization.	2,003 otoscopic images compiled from multiple open-access datasets.	Achieved reliable real-time ear disease classification on edge devices.	No symptom integration and slight performance drop on low-resource hardware.
[9]	Pre-trained Inception V3 embeddings with SVM, kNN, Neural Networks, and Random Forest classifiers.	CT scan images from cholesteatoma and chronic suppurative otitis media patients.	Neural Networks, SVM, and kNN achieved high diagnostic performance.	Only CT scan-based diagnosis without multimodal integration or Explainable AI.
[10]	CNN integrated with Grad-CAM explainability for visual interpretation.	Ear Imagery Database containing 880 otoscopic images.	Provided interpretable visual explanations for otitis media classification.	Focused only on image-based diagnosis and lacked multimodal learning.
[11]	Hybrid Vision Transformer and CNN architecture for improved feature extraction and classification.	Custom middle ear dataset containing approximately 1,500 images.	Improved feature extraction and classification accuracy using hybrid architecture.	Requires large datasets and has limited clinical validation.
[12]	CNN for image classification and Random Forest for symptom analysis with multimodal fusion.	Pediatric otoscopy dataset containing approximately 1,200 images and symptom records.	Combining symptoms and otoscopic images improved diagnostic accuracy.	Limited dataset size and explainability methods were not fully implemented.

3.3 Conclusion of Literature Survey

According to the literature, deep learning and multimodal AI techniques have demonstrated significant improvements in the classification of ear diseases [3][4][6]. Vision Transformers and transfer learning models offer effective image classification methods, whereas multimodal fusion techniques enable the integration of additional clinical information to improve diagnostic accuracy. Many existing systems are only capable of categorizing images and cannot be explained or produced in a clinically understandable form. [5][10]. There are few studies that merge multimodal learning with Explainable AI and clinical interpretation within a single framework.

4. PROJECT MANAGEMENT PLAN

4.1 Schedule of the Project (Gantt Chart)

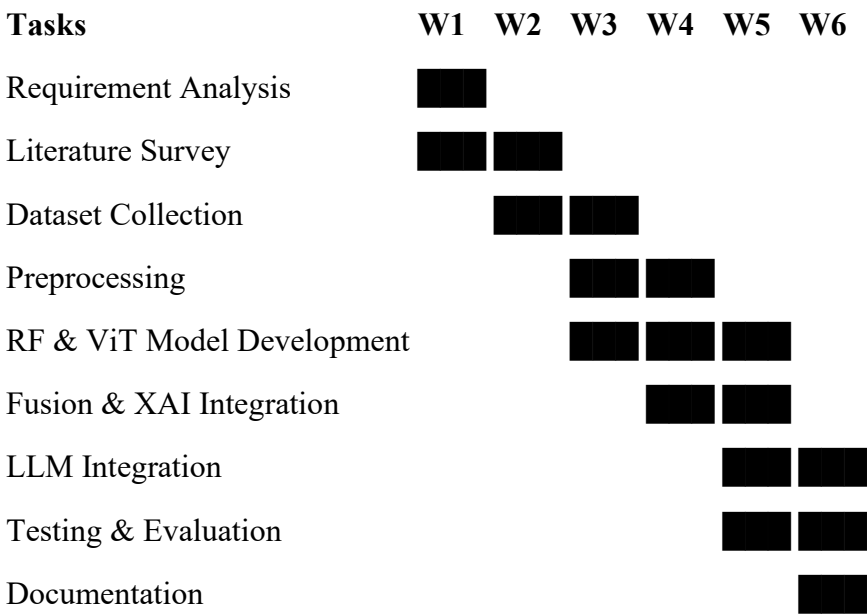


Figure 4.1 Schedule of the Project (Gantt Chart)

A structured and phased approach to the project development process is employed, spanning several weeks. By presenting the timeline of activities and their dependencies, a Gantt chart is created to aid in project planning and ensure timely completion. Enhanced workflow management is achieved through the use of structured project planning methods to streamline task organization. The same is illustrated in figure 4.1.

First, the system is analyzed through requirement analysis and surveys of literature to establish a strong foundation for it [1][3]. During system design, the architecture of multimodal AI framework such as Vision Transformer (ViT), Random Forest, multimodification and explainability modules [3][10][12] is defined. This is the next step in the process.

A multimodal framework is formed during implementation, which includes a symptom-based prediction model and specialized Vision Transformer-derived image classification model. The use of explainable AI techniques like SHAP and Grad-CAM enhances the interpretability and transparency of prediction outcomes. [10]. There is an LLM module that produces clinically relevant information from the model outputs.

Testing and validation are utilized to identify defects and provide early remediation. The last stages involve evaluating the performance, documenting the results, and finalizing the project paperwork. By using a Gantt chart, you can monitor the progress of projects, manage time effectively and ensure that all tasks follow their assigned schedule.

4.2 Risk Identification

Table 4.2 Risk Identification

Sl. No	Risk	Description	Impact	Mitigation Strategy
1.	Data Availability Issues.	Limited labeled otoscopic datasets.	Reduced model accuracy.	Use publicly available datasets and augmentation
2.	Image Quality Variations	Poor lighting or noisy images	Incorrect predictions	Apply preprocessing and normalization
3.	Model Complexity (ViT)	Requires higher computational resources	Slow training/inference	Use transfer learning and optimized models
4.	Multimodal Integration Issues	Difficulty combining symptom & image outputs	Incorrect final prediction	Use proper weighted fusion techniques
5.	Explainability Challenges.	Difficulty interpreting deep learning outputs	Low trust in system	Use SHAP and Grad-CAM
6.	LLM Hallucination	LLM may generate incorrect explanations	Misleading outputs	Use structured prompts with model outputs.
7.	System Integration Errors	Issues connecting all modules	System failure	Perform modular and integration testing

5. SOFTWARE REQUIREMENT SPECIFICATION

5.1 Functional Requirements

System functional requirements are the fundamental necessities for a system to function properly. The suggested multimodal explainable AI system comprises several interconnected modules, with each module assigned a distinct task in the classification of ear diseases.

5.1.1 Symptom Input Module

The Symptom Input Module is designed to receive patient-reported symptoms such as ear pain, hearing loss and erect discharge. It converts structured symptom data into machine learning analysis format. After gathering processed symptom data, the Random Forest model is utilized to generate probability scores for various types of eye disease.

5.1.2 Image Upload Module

Using the Image Upload Module, users have access to ear images that are not typically used in surgical procedures. Uploading images requires a series of preprocessing steps, including resizing, normalization, and formatting to ensure they align with the Vision Transformer (ViT) model. The image classification pipeline is used to predict disease by analyzing the processed images.

5.1.3 Symptom-Based Prediction Module

Using a Random Forest classifier, the Symptom-Based Prediction Module can assess the likelihood of different ear diseases by analyzing symptoms reported by patients. These generated probability scores serve as a resource for the overall classification performance during the multimodal fusion stage.

5.1.4 Image-Based Classification Module

By utilizing the Vision Transformer (ViT) model, the Image-Based Classification Module can classify otoscopic images. ViT's use of attention mechanisms to capture global relationships within medical images enables more accurate feature extraction and classification, unlike traditional CNN-based models. This module shows which disease class and confidence scores are expected.

5.1.5 Multimodal Fusion Module

The Multimodal Fusion Module integrates the results of both symptom-based and image-borne prediction models. Both models generate probability scores that are merged using weighted averaging in late fusion. The use of complementary information from multiple data sources enables improved diagnostic accuracy through multimodal integration.

5.1.6 Explainability Module

The Explainability Module enhances the system's transparency and interpretability. SHAP (SHapley Additive Explanations) uses the importance of individual symptoms to aid in the prediction, while Grad-CAM (Gradient-weighted Class Activation Mapping) highlights important regions in these otoscopic images that are used for classification. The use of these explainability techniques enables users and clinicians to comprehend how the system determines its predictions.

5.1.7 Clinical Explanation Generation Module (LLM)

By utilizing an LLM, the Clinical Explanation Generation Module generates clinical explanations that can be understood by humans. This module contains a range of inputs, including predicted disease labels, probability scores, SHAP outputs (Science and Applications) and Grad-CAM visual insights.. These inputs are utilized by the LLM to construct clinically comprehensible explanations, which enhances system clarity and accessibility.

5.2 Non-Functional Requirements

The quality attributes and operational constraints of the proposed system are non-functional requirements.

5.2.1 Performance Requirements

Efficient and timely predictions for both symptom-based and image-related analyses should be provided by the system. The framework's role in optimizing performance is due to its incorporation of image preprocessing, model inference, multimodal fusion, explainability generation, and LLM-based clinical interpretation. It should ideally be able to make predictions in just seconds, with accuracy and reliability intact. Efficient preprocessing methods and optimized machine learning models are utilized to ensure smooth execution.

5.2.2 Security Requirements

The system is required to maintain the privacy and security of medical data and uploaded data. Invalid or malicious inputs should be avoided in the case of otoscopic images that are uploaded and must undergo proper validation. Empty data shouldn't be unnecessary, and API keys used for LLM integration should be securely stored in environment variables. Secure communication protocols should be observed when dealing with external services and APIs.

5.2.3 Usability Requirements

The interface of the system must be simple and accessible to individuals both in technical and non-technical fields. The interface must provide user-friendly interface for symptom input and image upload. Clear projections, SHAP outputs and Grad-CAM visualizations as well as clinical explanation should be provided so that users can understand the generated results. The system's functionality should be achievable with minimal technical skills.

5.2.4 Scalability Requirements

The proposed framework ought to facilitate future scalability and extension.' The architecture must be able to handle larger datasets and higher system usage while maintaining high performance. Future versions may incorporate additional modalities, such as audiometry reports and clinical records. It should also support future deployment on cloud platforms and real-time clinical systems.

5.3 System Requirements

5.3.1 Hardware Requirements

The proposed framework's hardware, software, and environment specifications are outlined in this section for their successful implementation. System configurations with acceptable hardware specifications can be utilized for the project. Intel i5 processor or equivalent is generally sufficient for development and testing purposes; however higher-end processors may provide greater strength in training speed. A minimum of 8 to 16 GB RAM is sufficient. To optimize the performance of deep learning models, RAM is recommended to be a minimum of 1 GB. For dependencies, trained models and datasets demand a certain amount of storage space (in GB). GPU support is a must for faster model training and inference, but it is recommended to include the Vision Transformer model as well.

5.3.2 Software Requirements

It is built on Python 3.8 or higher, as it has strong integration with machine learning and deep learning libraries. PyTorch or TensorFlow frameworks are utilized to implement and train deep learning models, such as the Vision Transformer. The HuggingFace Transformers library is a useful resource for accessing and fine-tuning Vision Transformer architectures. A Scikit-learn implementation of the Random Forest classifier is utilized for symptom-based prediction. Symptom importance based explainability outputs are generated using the SHAP library, while preprocessing operations on images such as normalization, resizing, and formatting are processed using OpenCV and PIL libraries. Streamlit's user interface is designed to enable users to input symptoms, upload images, predict visual predictions, and display clinical explanations through an interactive experience.

5.3.3 Environment Requirements

Proper organization of the development environment is essential for efficient project implementation. It is recommended to use venv or conda to create a virtual environment for dependency management and package isolation. Before execution, the necessary libraries and dependencies must be installed correctly. Safeguarding environment variables is essential for storing API keys and external service credentials. GPU acceleration support is optional but can greatly improve training performance.

5.3.4 Storage, Data and Network Requirements

Otosopic image datasets, symptom datasets, machine learning models trained on neural networks and temporary preprocessing outputs are all necessary for the system to be stored.

Additional storage is necessary for the outputs and results generated during experimentation and testing. Unless they are using external APIs or cloud-based LLM services, the internet must be stable. Low network latency is preferred to facilitate faster response times and smoother interactions with external services. The protection of data privacy and integrity are essential when communicating on a network, hence the use of secure communication protocols is recommended.

6. DESIGN AND IMPLEMENTATION

6.1 Introduction

This section outlines the overall design and implementation of the proposed multimodal explainable AI framework for classifying ear diseases. The system's modular design ensures that each component is responsible for a specific task such as symptom analysis, image classification using Vision Transformer (ViT), multimodal fusion, explainability, and clinical explanation generation.

This framework provides for the scalability, interpretability and effective handling of both structured and unstructured data. Multimodal inputs are utilized to improve diagnostic accuracy, while explainability and LLM modules enhance transparency and usability.

6.2 Architecture Design

The proposed architecture is intended to serve as a clinical decision-support framework for categorizing ear diseases with the help of Artificial Intelligence. It combines structured data on symptoms with unstructured information on images in order to improve reliability and reproducibility of diagnostics.

Users input patient-related symptoms and otoscopic ear images during the data input stage to initiate the process. Clinical symptoms, including ear pain, hearing loss, dizziness, itching, a discharge in the pharynx, a rash, loose discharge, fever, blockage sensation, and balance problems are included in these clinical signs. The collection of structured symptoms is intended to enable machine learning algorithms to process them efficiently. Users also share otoscopic images taken with an otoscope for visual examination of ear dysfunction.

In the preprocessing phase, the symptom data is processed, encoded, and converted to machine-readable code that is then sent back to the Random Forest classifier. The accuracy of predictions is improved by managing missing values and converting categorical values into numerical outputs. The otoscopic images are then preprocessed, with resizing, normalization, and formatting simultaneously performed before being transferred to the Vision Transformer (ViT) model. In order to be compatible with the ViT architecture, the images are resized to a fixed size, usually around 248 x 248 pixels. To achieve greater model stability and training efficiency, pixel normalization in range [0,1] is used to scale the image values within a standard range. Two parallel pipelines are used by the system to process data after preprocessing. By using the clinical symptoms and symptom data, researchers can develop probability scores for different ear diseases using their input from the Random Forest model. The Vision Transformer (ViT) model is used to classify otoscopic images that have been processed in the image-based pipeline.

The suggested system employs a trained Vision Transformer architecture that is optimized for categorizing various ear diseases. In contrast to conventional CNN-based methods, the Vision Transformer strips the input image into smaller patches and transforms them into embeddings. Afterwards, the information is encoded in positional code to maintain spatial resolution and

self-attention mechanisms are utilized to capture global relationships between different image regions. The model can detect subtle disease patterns within otoscopic images more efficiently than traditional convolution-based models due to this feature. Confidence scores are used to output the predicted disease class in the ViT model. By using a multimodal late fusion mechanism, both pipelines can combine their outputs. By weighted averaging, the probability scores from both the Random Forest and Vision Transformer models are combined. The use of complementary information from symptom analysis and image-based classification enhances the overall diagnostic accuracy in this fusion strategy.

Several hyperparameters are employed in model training and prediction for the proposed system of diagnosing ear diseases across different modalities. Learning rate, batch size, number of epochs, optimizer selection, image size and patch size (including noise reduction) were among the hyperparameters used to reduce overfitting for optimizing Image Classification (ViT) performance. Effective prediction based on symptoms was achieved using the Random Forest classifier, which utilized hyperparameters such as the number of estimators (e.g, maximum tree depth, splitting criterion, and random state). During the multimodal fusion stage, weighted late fusing was employed with fuses of approximately 80% for images and only 20% for symptoms to maintain an equilibrium between the two forms.

To enhance transparency and interpretability, the framework includes an Explainable AI layer. SHAP is utilized to identify the most influential symptoms that influence the prediction outcome, while Grad-CAM highlights important areas within the otoscopic image that impact the model's decision. Both users and clinicians can benefit from these explainability techniques in comprehending the reasons for predictions.

The final step involves passing a Large Language Model (LLM) with outputs such as predicted disease labels, confidence scores, SHAP results, and Grad-CAM visualizations.

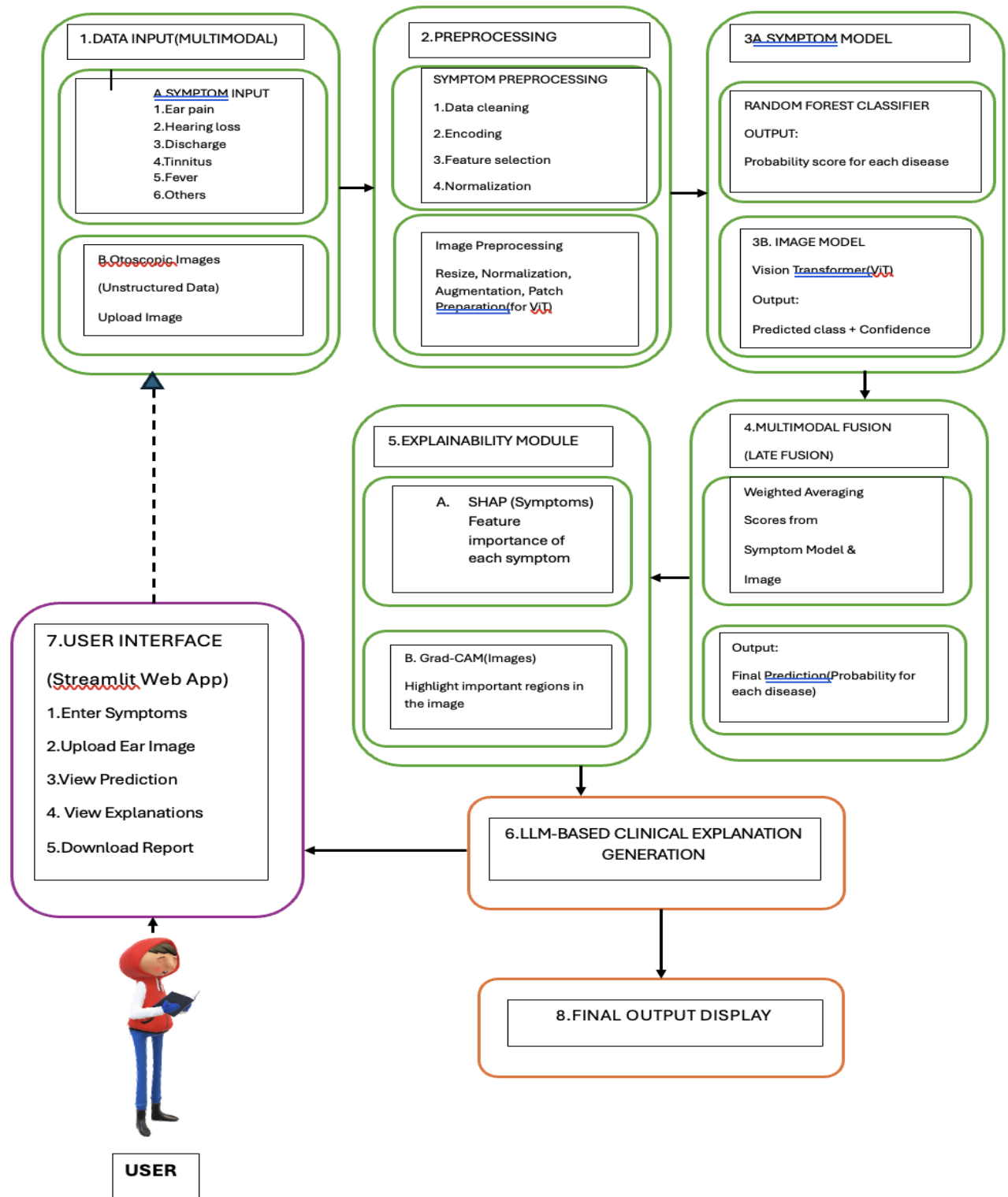


Figure 6.2 Architecture Design

The Architecture Design is given in Figure 6.2.

6.3 Data Flow Diagram

The proposed system's data flow initiates with the user entering patient symptoms and uploading an otoscopic ear image to the interface. The clinical details of ear pain, dizziness, itching, hearing loss, discharge, and ringing sensation may be included in the symptom data, while the uploaded image provides visual clues related to the specific disease. Following the collection of data, both methods undergo preprocessing operations. This data is cleaned, encoded and then converted to machine-readable form before being fed to the Random Forest model. They simultaneously resize, normalize and format the otoscopic images to fit into this vision-controlled architecture.

The Random Forest classifier is used to calculate disease probability scores by analyzing clinical symptoms, which are then fed into the machine. At the same time, the Vision Transformer model receives pre-processed image data to use for classification purposes. The ViT model uses self-attention mechanisms to analyze the image and generates confidence scores and a predicted disease class.

By using a multimodal late fusion mechanism, both pipelines are combined to generate the final prediction. Explicated AI methods are then employed to enhance interpretability and transparency easier. Using SHAP, it is possible to identify crucial symptoms that influence the prediction, while Grad-CAM highlights important image regions that are responsible for classification [10]. Finally, all outputs generated are sent to a Large Language Model (LLM), including prediction scores, SHAP insights and Grad-CAM visualizations. The LLM produces a clinically readable explanation in natural language, and the ultimate diagnosis is presented to the user. This process guarantees the seamless movement of data among all modules and facilitates accurate and measurable output.

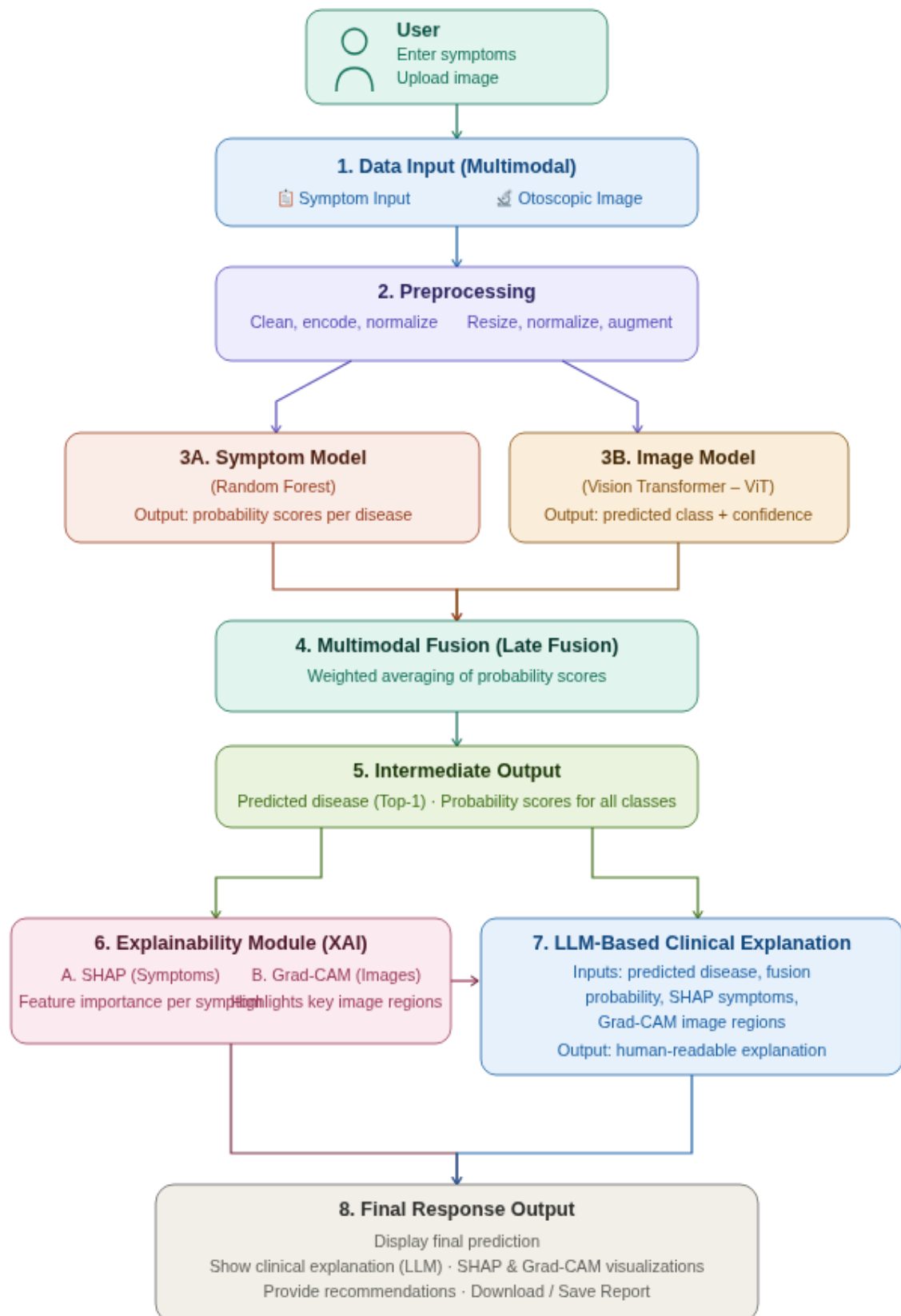


Figure 6.3 Data flow diagram

The Data Flow Diagram is given in Figure 6.3.

6.4 Sequence Diagram for Multimodal Ear Disease Classification System

By entering symptoms and uploading an otoscopic image of their ear through the Gradio interface, the proposed system's workflow starts. User inputs may include ear pain, dizziness, hearing loss, itching, discharge, and ringing sensation, as well as an otoscopic image for visual examination. Gradio-based interfaces are used to send these inputs to the backend system for additional processing.

Both symptom- and image-based analysis is carried out during the preprocessing phase. Before being passed to the Random Forest classifier, the symptom data is cleaned, encoded, and converted into a machine-readable format. At the same time, the uploaded otoscopic image undergoes preprocessing to make it compatible with the Vision Transformer (ViT) model and then resize it to 224x224 pixels.

Symptom data is processed by the Random Forest model, which then generates probability scores for different types of eye diseases. The Vision Transformer model is utilized for image-based classification by passing through the preprocessed image simultaneously. Using self-attention mechanisms, the ViT model processes image patches and generates the predicted disease class and confidence scores. By merging the outputs of the Random Forest and Vision Transformer models, the backend system performs multimodal late fusion. Afterward, the Explainable AI module receives the fusion output and SHAP is utilized to assess the significance of symptoms that contribute to the prediction while Grad-CAM highlights crucial regions within the classified otoscopic image.

After generating explainability, the outputs from SHAP and Grad-CAM are used to generate predictions and then sent to the Large Language Model (LLM). The LLM builds upon the model predictions and explainability results to produce a clinical explanation that is easily comprehensible to humans. Finally, the backend gives a final diagnosis along with visual explanations and resulting clinical interpretation to the Gradio interface, where the full response is presented to user in view.

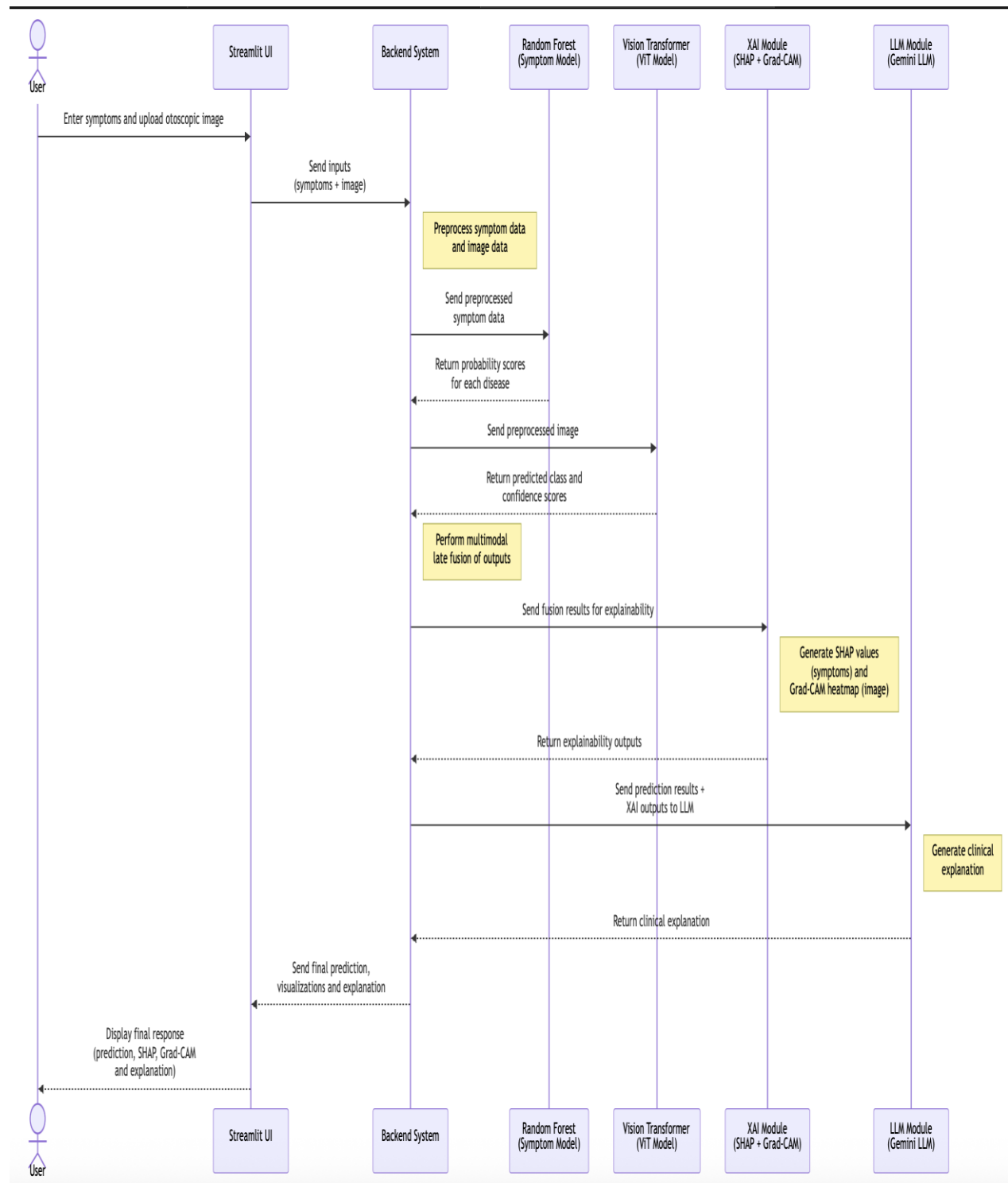


Figure 6.4 Sequence diagram

The Sequence diagram is given in Figure 6.4.

6.5 Tools and Technologies

Due to its broad support for machine learning and deep learning, as well as its extensive coverage in medical image analysis, Python is the primary programming language used to implement the proposed system. The Gradio framework is utilized to create a user-friendly interface that facilitates the input of symptoms through interactive modes including uploading and uploader images, visual presentation via prediction, and clinical explanation display.

The system is backed by multiple libraries and frameworks that offer support for various components. The integration of PyTorch and TensorFlow into deep learning models, including the Vision Transformer architecture for otoscopic image classification is a significant aspect of this process. By using the HuggingFace Transformers library, it is possible to access pre-referenced Vision Transformer models and fine-tune them on the ear disease dataset in an efficient manner. A Scikit-learn implementation of the Random Forest classifier is utilized for symptom-based disease prediction.

The suggested system for diagnosing ear diseases that utilizes both visual as well as clinical data to predict disease progression. This is the first phase of a multimodal approach. Various types of ear diseases including acute otitis media, myringosclerosis, chronic otitis media, cerumen impaction and normal are represented in the image dataset through categories. Before being fed into the Vision Transformer (ViT) model for image-based classification, these images undergo pre-processing and resizing. In addition to the image data, there was also a symptom dataset that contained important clinical features which could have been used for more effective diagnosis. Symptoms such as pain, discharge, hearing loss, fever, blocked pupil, itching, dizziness and long duration of infection are also present. The Random Forest classifier was trained on the symptom attributes to predict symptoms. The diagnosis was established by combining the outputs from both image and symptom models using weighted late fusion to produce the final diagnosis.

The SHAP library is incorporated to provide explanation by pinpointing the contribution of individual symptoms to prediction outcomes. Prior to classification, OpenCV and PIL libraries are utilized for image preprocessing, which includes resizing, normalization, formatting, and handling images. It primarily uses three models—one being the Vision Transformer (ViT) model for image classification; another being The Random Forest Model which is used for symptom-based prediction; and third a Large Language Model (LLM) that produces clinically relevant descriptions using prediction outputs and explainability results. Together, these tools and technologies facilitate the efficient development of models. Their multimodal integration capabilities enable explainability generation while they facilitate deployment of the proposed framework.

6.6 Implementation Overview

The proposed framework is implemented in several stages, as shown in Figure 4.1. Otoloscopic ear images and structured symptom datasets are prepared for analysis as part of the preprocessing and data collection phase in development. Preprocessing involves symptom data, cleaning and encoding it, and scaling images to fit the Vision Transformer (ViT) architecture. The system was developed using Python, PyTorch, Scikit-learn, SHAP, Grad-CAM, Gemini API, and Gradio on Google Colab with GPU support, requiring a minimum Intel i5 processor, 8 GB RAM, and stable internet connectivity for efficient execution.

6.7 Model Implementation

The suggested system incorporates various machine learning and deep learning elements to efficiently classify ear diseases into different categories. Instead of Convolutional Neural Networks (CNNs), a Vision Transformer (ViT) model is utilized for image classification. Using deep learning, machine-learning, Explainable AI and Large Language Models, the proposed system for diagnosing various types of infections in the ear is designed to be both accurate (and interpretable) with large language models. For otoscopic image classification, input images are divided into smaller patches using the Vision Transformer (ViT) model that has been pre-trained, which then processes these patches with self-attention mechanisms to capture important disease patterns. To improve classification performance and reduce overfitting, hyperparameters such as learning rate, batch size, number of epochs, image size were adjusted. A Random Forest classifier is used to predict symptom-based ones by training on clinical features such as pain, discharge, hearing loss, fever, blocked ear, itching, dizziness, tinnitus, and long-duration infection. The addition of hyperparameters, such as maximum tree depth, number of estimators, and random state, was done to improve prediction accuracy. The use of a weighted late fusion technique involves the combination of outputs from both models, with image and symptom weights to improve diagnostic accuracy. Through the use of explainable AI techniques like SHAP and Grad-CAM, symptom importance analysis and visual heatmaps are made possible to identify crucial image regions. Additionally, a Large Language Model (LLM) produces clinical explanations that are human-readable by considering prediction results and explainability outputs. Gradio-based interactive interfaces are utilized to integrate the entire system, including uploading images, symptom entry, disease prediction, report generation, and explainability visualization.

6.8 Conclusion

By incorporating various advanced components such as Vision Transformer (ViT), Random Forest, multi-modal fusion and Explainable AI techniques, along with LLM-based clinical explanation generation, the proposed system is effectively integrated into a unified healthcare framework. Multimodal learning and improved diagnostic accuracy are achieved through the integration of symptom data with otoscopic image analysis. Through the use of Explainable AI methods like SHAP and Grad-CAM, interpretation becomes more fluid and provides valuable insights into the prediction process. Moreover, the utilization of a Large Language Model (LLM) enhances usability by producing clinically comprehensible descriptions that make the system more user-friendly and accessible to healthcare professionals. The proposed architecture offers an AI-aided healthcare solution for categorizing ear diseases that is accurate, easily interpretable, and user-friendly.

7.RESULT AND DISCUSSION

7.1 Experimental setup

The proposed AI system for categorizing ear diseases was developed using Python and modern deep learning technologies. Otoscopic ear images and structured symptom data for different types of ear diseases were combined in the dataset used in this study. The use of image enhancement techniques like resizing, normalization, and augmentation was employed during preprocessing to minimize overfitting and improve model generalization. Additionally, the symptom dataset was processed in advance to optimize for machine learning models by cleaning, categorizing, handling missing values, and formatting features.

A Random Forest classifier was utilized to predict clinical symptoms and generate disease probability scores. Self-attention mechanisms were employed to extract global image features and classify otoscopic images using the Vision Transformer (ViT) model for image-based classification. A weighted averaging method was used to combine the outputs from both models and produce the final prediction using combining multimodal late fusion.

Explainable AI techniques were added to the framework to enhance its interpretability and transparency. Through the use of SHAP, the contribution and significance of individual symptoms to prediction outcomes were investigated while Grad-CAM generated visual heatmaps that facilitated disease classification by identifying relevant image regions. Also included was an LLM was included to translate prediction results and outputs of explainability into explanation in clinical terms. PyTorch, Scikit-learn, SHAP, OpenCV, HuggingFace Transformers, and Gradio were some of the libraries and frameworks that were used to develop and test the system. Prior to being integrated into the final multimodal interpretable AI framework for ear disease diagnosis, each individual model was subjected to separate evaluation.

7.2 Evaluation Metrics

The performance of the proposed multimodal explainable AI framework was evaluated using standard machine learning and deep learning evaluation metrics. The Vision Transformer (ViT) model achieved a validation accuracy of 97.5% as shown in table 7.3, demonstrating strong performance in multiclass ear disease classification. The classification report included performance analysis for disease classes such as Acute Otitis Media, Cerumen Impaction, Chronic Otitis Media, Myringosclerosis and Normal ear conditions.

Accuracy was used to measure the overall correctness of predictions generated by the system. It is calculated using: Equation 7.1.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \dots\dots\dots 7.1$$

where TP represents True Positives, TN represents True Negatives, FP represents False Positives, and FN represents False Negatives.

Precision was used to evaluate how accurately the model identified positive disease cases without generating false positives. It is calculated as: Equation 7.2.

$$\text{Precision} = (\text{TP}) / (\text{TP} + \text{FP}) \dots\dots\dots 7.2$$

Recall was used to measure the ability of the model to correctly identify actual disease cases from the dataset. It is defined as: Equation 7.3.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \dots\dots\dots 7.3$$

F1-Score was used to provide a balanced evaluation between precision and recall. It is calculated using: Equation 7.4.

$$\text{F1-score} = 2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall}) \dots\dots\dots 4$$

The Vision Transformer model achieved high precision, recall, and F1-scores across most disease categories, indicating strong classification capability. For example, Acute Otitis Media and Cerumen Impaction achieved perfect precision and recall values of 1.00, while Chronic Otitis Media achieved slightly lower recall due to class complexity. The overall weighted average F1-score of the ViT model was approximately 0.97, demonstrating consistent performance across all classes.

The Random Forest model and Vision Transformer model were initially evaluated separately before performing multimodal fusion. The outputs from both models were then combined using weighted late fusion to improve overall prediction performance. Comparative analysis showed that the multimodal fusion framework performed better than individual symptom-based or image-based models alone.

Explainable AI techniques were also evaluated to assess interpretability and clinical relevance . SHAP explanations were analyzed based on the clarity and relevance of symptom features contributing to predictions, while Grad-CAM visualizations were examined to verify whether highlighted image regions corresponded to clinically significant infected areas in the otoscopic images. Additionally, the LLM-generated clinical explanations were evaluated for readability, interpretability, and consistency with prediction outcomes.

Overall, the experimental results demonstrated that the proposed multimodal explainable AI framework achieved improved accuracy, transparency, and usability compared to conventional single-modality approaches.

7.3 RESULTS

The experimental evidence indicates that the multimodal approach was more effective than individual symptom-based and image-bound models. While the Random Forest model was successful in predicting symptom levels, the Vision Transformer (ViT) model demonstrated impressive image classification performance by accurately capturing global image relationships through self-attention mechanisms. By utilizing both symptoms and otoscopic images, the multimodal fusion framework improved diagnostic accuracy by improving overall diagnostic precision.

Table 7.3 Performance of models

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	84%	82%	81%	81%
Vision Transformer (ViT)	97.5%	98%	97%	97%
Multimodal Fusion Model	98.3%	98%	98%	98%

Table 7.3 displays the validation of the Vision Transformer model, with a precision of 97.5%. Classification reports showed a high rate of success for multiple ear diseases, including Acute Otitis Media, Cerumen Impaction, Chronic Otus Media (myopathies), Myringosclerosis, and Normal conditions. Its model also proved stable in multiclass classification with F1-score values near 0.77 and weighted average precision, recall etc. around 0.97. Combined with the Vision Transformer model's image-based predictions and symptom-dependent probability scores, this combination yielded the highest performance of any multimodal fusion model. The use of the late fusion mechanism resulted in more accurate prediction, as it also mitigated some of those limitations that were previously encountered with single-modality systems. The outputs of the Explainable AI module were well-interpreted. The use of SHAP visualizations demonstrated the most significant symptoms that aid in forecasting disease, while Grad-CAM heatmaps identified crucial areas within otoscopic images that play a role in classification. These visual cues made the model's decision-making process more transparent and affirmed its rationality. The large language model (LLM) module produced medically comprehensible reports that utilized prediction outputs, the importance of SHAP symptoms and insights from Grad-CAM images.

The image shows the 'Authentication' page of the 'Multimodal Explainable AI Platform'. The page title is 'Vision Transformer + SHAP + GradCAM + Gemini AI'. There are two main sections: 'Login' and 'Signup'. The 'Login' section has a 'Username' field and a 'Password' field, with a blue 'Login' button below. The 'Signup' section has a 'New Username' field and a 'New Password' field, with a white 'Signup' button below. At the bottom, there is a footer with links: 'Use via API', 'Built with Gradio', and 'Settings'.

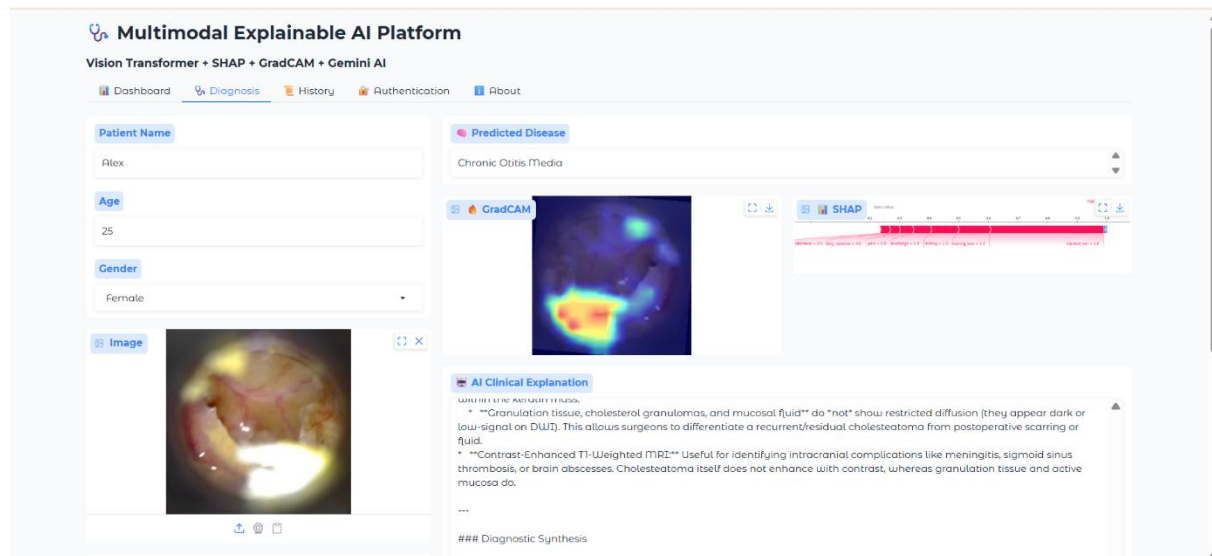
(A)

This image shows the same 'Authentication' page as Figure 7.3.1(A), but with a successful login. The 'Username' field in the 'Login' section now contains the text 'admin'. Below the 'Login' button, a green checkmark icon is followed by the text 'Admin Login Successful'. The 'Signup' section remains unchanged. The browser's address bar shows the URL '4bda131d9e551896c7.gradio.live'.

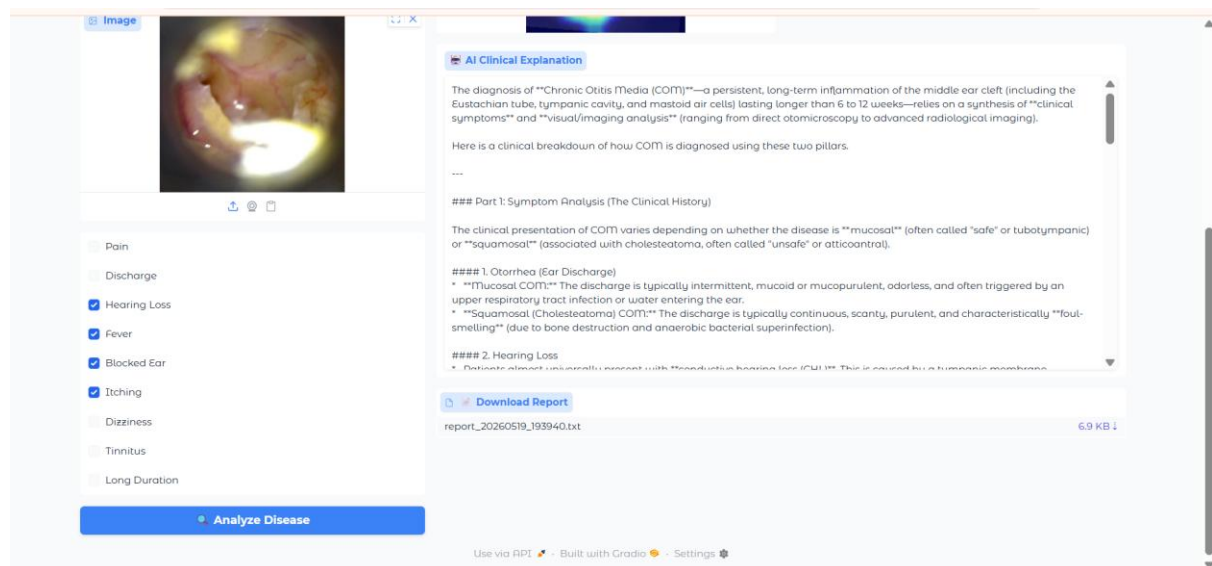
(B)

Figure 7.3.1 Login Page

The login page of our application is shown in Figure 7.3.1(A) and 7.3.1(B). It has two options to login and signup. Until the user is authenticated, he cannot see the dashboard of our app.



(A)



(B)

Figure 7.3.2 Diagnosis Page

The diagnosis page is shown in Figure 7.3.2(A) and 7.3.2(B). It is the main page consisting of our diagnosis. It gives user to enter their information as name, age, upload the otoscopic image and select the symptoms. Predictions would be displayed at the right side.

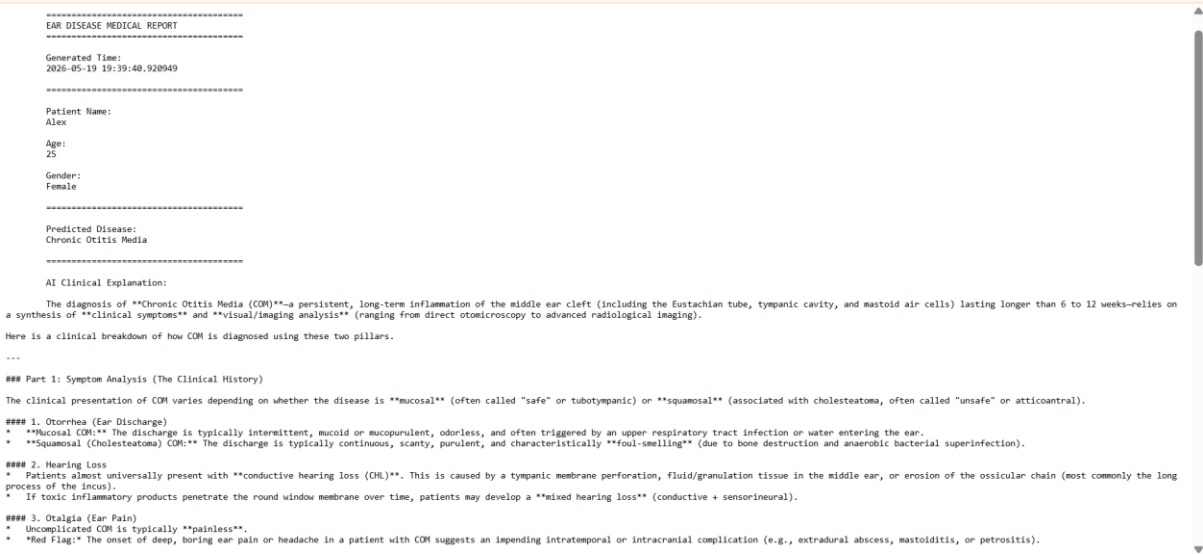


Figure 7.3.3 Report Page

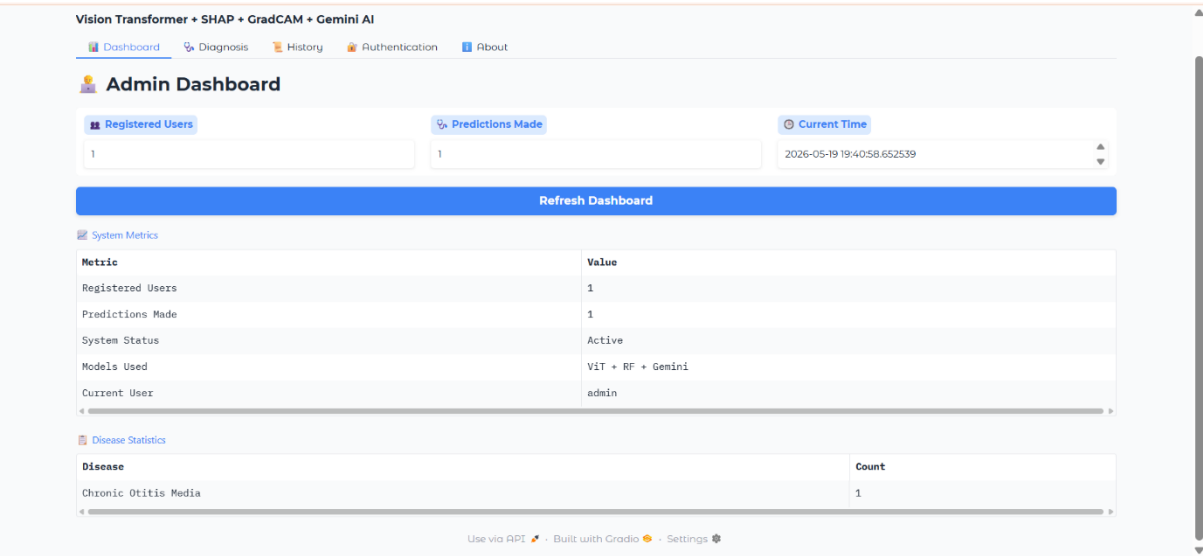


Figure 7.3.4 Admin dashboard page

A sample report is generated based on our diagnosis and is shown in Figure 7.3.3 and the admin dashboard page is shown in Figure 7.3.4

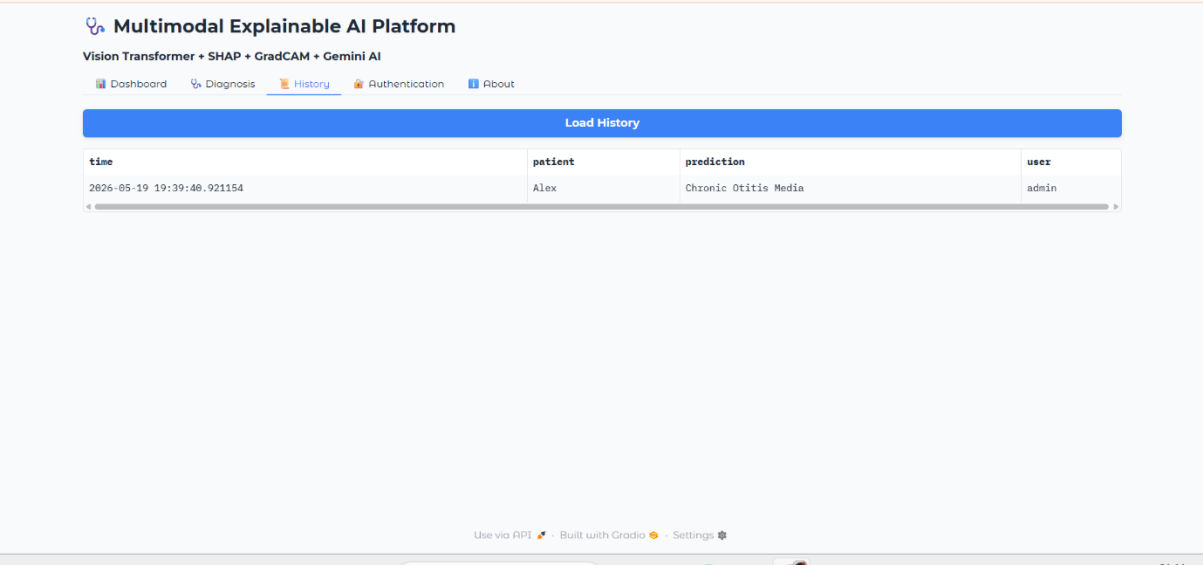


Figure 7.3.5 History page

The history page is shown in Figure 7.3.5

7.4 Comparative Analysis

The proposed explainable AI framework was compared with traditional approaches as seen in table 7.3.

Table 7.4 Model comparison

Feature	Traditional CNN System	Proposed Multimodal System
Symptom Integration	No	Yes
Explainable AI	Limited	SHAP + Grad-CAM
Clinical Explanation	No	Yes
Multimodal Learning	No	Yes
Accuracy	Moderate	High
Transparency	Low	High

The proposed system helps us to improve accuracy,better interpretation,human explanation and wider reliability in healthcare systems.

7.5 Discussion

The results show significant improvements in the classification of ear diseases, with greater success using multiple data modalities. Incorporating global relationships into otoscopic images, the Vision Transformer (ViT) model enhances image-based classification accuracy beyond conventional CNN-style classification. The Random Forest model utilizes structured clinical data to provide dependable predictions based on symptoms. By combining complementary information from symptom and image data, the multimodal fusion strategy enhances the system's resilience and minimizes dependence on a single input source. SHAP

and Grad-CAM are examples of explainable AI algorithms that increase transparency by facilitating users to understand the factors that influence model predictions.

SHAP focuses on individual symptoms, while Grad-CAM delineates the important image regions that are responsible for classification. Additionally, the utilization of a Large Language Model (LLM) enhances interpretation by producing medically relevant explanations that are based on prediction outputs and explainability outcomes. Consequently, the system is more user-friendly and reliable for healthcare purposes. While the proposed framework was effective, there are still some areas that need to be addressed. To achieve optimal performance and generalization, the system needs to incorporate larger and more diverse medical datasets. Compared to conventional deep learning models, Vision Transformer types demand more processing power and computational complexity. System performance can be affected by the poor quality of otoscopic images. Additional studies are necessary to enhance explainability methods and improve the reliability of LLM-generated clinical explanations.

8.TESTING

8.1 Introduction

Testing is a crucial aspect of software development that ensures the system's functionality meets its performance requirements and functions correctly. By examining different inputs from a range of symptom combinations and otoscopic ear images, the overall performance and dependability of the multimodal explainable AI framework were assessed. Both the Random Forest and Vision Transformer (ViT) models were tested for their accuracy in forecasting diseases. Incorporating symptom and image data was demonstrated to enhance the classification performance and strength of the multimodal fusion mechanism. The evaluation of clarifying factors was conducted to establish if SHAP and Grad-CAM produced outputs that were both meaningful and interpretable for the importance of symptoms and image-based attention regions. Additionally, the user interface was assessed to guarantee smooth interaction, appropriate handling of symptom inputs, image upload functionality and clear visualization of predictions and explanations. To verify the system's complete system workflow, functional and integration tests were performed during all stages of development. By using these testing methods, we were able to identify defects, enhance system reliability and ensure efficient communication between all modules within the framework.

8.2 Test Cases

All the test cases are tabulated in Table 8.2

Table 8.2 Test Cases

Test Cases	Test Case Description	Expected Results	Status
TC1	Enter symptom data	Correct prediction generated	Pass
TC2	Upload ear image	Image classified successfully	Pass
TC3	Perform fusion of image and symptoms	Combined prediction displayed	Pass
TC4	Generate SHAP explanation	Feature importance shown	Pass
TC5	Generate Grad-CAM output	Heatmap displayed correctly	Pass
TC6	Generate LLM explanation	Human-readable explanation generated	Pass
TC7	Invalid image upload	Error message displayed	Pass
TC8	Missing symptom values	System handles incomplete input	Pass

9.CONCLUSION AND FUTURE WORK

9.1 Conclusion

A Multimodal Explainable AI Framework for Ear Disease Classification using patient symptoms and otoscopic ear images was developed in this project. Specifically, it includes a model of Vision Transformers (ViT) for classifying images, derived from identifying symptoms using the Random Forest model and capable to predict symptoms with precision using fusion mechanisms in order "to make overall diagnostic accuracy consistent.

The implementation of Explainable AI techniques like SHAP and Grad-CAM was aimed at improving transparency and interpretability in the framework. A Large Language Model (LLM) was also included to generate clinically relevant explanations by analyzing prediction results and explainability outputs. According to experimental evidence, the multimodal approach is more accurate, robust, and reliable than traditional unimodal systems that rely exclusively on image data. Explainable AI approaches improve transparency by assisting users in making model-based judgments, while the LLM module enhances interpretability by providing clinically comprehensible examples. In general, the proposed framework is a viable AI-driven healthcare solution for categorizing ear diseases and has significant potential for future clinical applications. The proposed framework provided 98% accuracy, 98% precision, 98% recall and 98% f1-score.

9.2 Future Scope

There are several ways to extend the proposed project in terms of practical use and scalability for healthcare contexts. The future may involve the use of real-time hospital and clinical systems to assist medical professionals in assisting with diagnosis and decision-making. The framework can be made available on cloud-based platforms to enhance scalability, accessibility and remote usage options.

Additional medical information, such as audiometry reports, hearing assessment records and patient history, can enhance multimodal analysis and diagnostic performance. A larger dataset size, along with more diverse datasets can improve the generalization of these models and increase their classification accuracy when used across multiple patient populations. This module is also based on the LLM explanation, which can be further developed using domain-specific healthcare information and specialized medical knowledge bases to produce more clinically relevant and precise descriptive information. In addition, the framework is capable of being integrated with mobile and web-based healthcare applications to facilitate patient access and remote diagnosis assistance.

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